Vector Spaces IV: Quotient Spaces and Counting over Finite Fields

Problem sheet for the Day 5 class · CSIR NET / GATE / IIT JAM

Name: Date: Score: / 10

Instructions. 10 problems from CSIR NET, GATE (MA) and IIT JAM. A tag with an exam and year marks an actual previous-year question; a tag reading *Exam pattern* marks a problem written in the same style. MCQ: one correct option. MSQ: one or more correct options. NAT: write the number. Suggested time: 30 minutes. **The answer key is posted 24 hours after this sheet.**

1. Exam pattern NAT Let $V = M_5(\mathbb{R})$ and W the subspace of symmetric matrices. Then $\dim(V/W) =$ _____.

2. Exam pattern NAT Let W be a 2-dimensional subspace of $\mathbb{F}_3^4$. The number of elements of $\mathbb{F}_3^4/W$ is _____.

3. Exam pattern NAT The number of invertible 2×2 matrices with entries in $\mathbb{F}_3$ is _____.

4. Exam pattern NAT The number of 2×2 matrices over $\mathbb{F}_3$ with determinant 1 is _____.

5. Exam pattern NAT The number of 1-dimensional subspaces of $\mathbb{F}_3^3$ is _____.

6. Exam pattern NAT The number of 2-dimensional subspaces of $\mathbb{F}_2^4$ is _____.

7. Exam pattern NAT The number of ordered bases of $\mathbb{F}_2^3$ is _____.

8. Exam pattern NAT The number of 2×2 matrices A over $\mathbb{F}_2$ with $A^2 = 0$ is _____.

9. Exam pattern MSQ Let V be the real polynomials of degree at most 3 and $W = \{p \in V : p(1) = 0\}$. Which of the following are true?

(A) $\dim(V/W) = 1$
(B) $p + W = q + W$ iff $p(1) = q(1)$
(C) $V/W \cong \mathbb{R}$
(D) $\dim W = 2$

10. Exam pattern NAT The number of subspaces W of $\mathbb{F}_2^3$ with $\dim(\mathbb{F}_2^3/W) = 1$ is _____.